

AHMED GHARIB

Principal Advanced Analytics Engineer

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PROFESSIONAL SUMMARY

Data Engineering and AI professional with 8+ years of experience building scalable data platforms, high-performance pipelines, and intelligent automation across Azure, Snowflake, AWS, GCP, and Databricks. Led teams in architecting Medallion-pattern warehouses with DBT and delivering async ETL pipelines with 68x performance gains. Advanced practitioner of Agentic AI tools (Claude Code, GitHub Copilot, N8N) with hands-on experience designing AI-powered pipelines using Azure OpenAI GPT and LangChain. Recognized internally for driving AI adoption through multiple training sessions on coding assistants and AI workflows for engineering teams. Known for measurable impact: 98% data accuracy via automated entity correction, 93% reduction in manual reporting effort, and end-to-end competitor intelligence systems built on Agentic AI.

CORE SKILLS

Cloud & Data Platforms: Azure (Synapse, ADF, Blob Storage, Key Vault, AKS, DevOps), Snowflake, AWS (Redshift), GCP (BigQuery), Databricks, Microsoft Fabric

Data Engineering: ETL/ELT Pipelines, DBT, SSIS, Delta Lake, Medallion Architecture, Data Modeling, Data Warehousing, Data Governance

Programming & Tools: Python, SQL, PySpark, Pandas, Async/Await, Selenium, Git, Docker, Kubernetes, N8N

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, Agentic AI, LLM Integration, Azure OpenAI GPT, LangChain, Claude Code, GitHub Copilot

BI & Visualization: Power BI, SSRS, Paginated Reports, DAX, Excel (Power Query, VBA)

Databases: SQL Server, PostgreSQL, MongoDB, Oracle, Azure SQL, Snowflake

PROFESSIONAL EXPERIENCE

Principal Advanced Analytics Engineer

ITWorx

Oct 2023 – Present

Cairo, Egypt (Hybrid)

- Spearheaded the ADX Digital Transformation for Abu Dhabi Securities Exchange (Core42), delivering data infrastructure modernization for a leading UAE financial institution serving investors, regulators, and exchange operators.
- Developed a Python CLI tool that automated Power BI Paginated Report standardization, cutting processing time from 4 hours to 15 minutes (93% reduction) across 190+ reports deployed to 4 environments (Dev, Test, UAT, Prod).
- Optimized Azure SQL Database read queries for the Market Watch system, significantly improving real-time data retrieval speeds for trading platform users.
- Refined table structures in Azure Synapse Dedicated SQL Pool Data Warehouse, ensuring efficient data processing and storage for the exchange platform.
- Designed and implemented a Power BI report for real-time performance monitoring of the Operational Data Store Database (ODSDB), enabling proactive system management.
- Led presales activities and built proof-of-concept Agentic AI solutions using N8N, Azure OpenAI GPT, and LangChain, positioning the team for AI-driven automation opportunities.
- Conducted multiple training sessions for ITWorx engineering teams on AI coding assistants (Claude Code, GitHub Copilot) and Agentic AI workflows, accelerating team-wide adoption of AI-augmented development practices.
- Mentored junior engineers on analytics best practices, driving adoption of modern data engineering and ML techniques across the analytics function.

Tech: Azure Synapse Analytics, Azure SQL, Power BI, Paginated Reports, PostgreSQL, Python, PowerShell, N8N, Agentic AI

Senior Data Engineer

Dec 2021 – Oct 2023

Raisa Energy LLC

Cairo, Egypt (Hybrid)

- Led a team of 4 engineers in migrating the data warehouse from a multi-layer architecture to a proper Medallion architecture (Bronze/Silver/Gold) using Snowflake and DBT, improving data lineage and governance.
- Designed and built the Entity Correction Model in DBT/Snowflake, an automated system that corrects revenue and expense allocations to proper legal entities using exact and fuzzy matching against Jarvis ownership data — improving entity attribution accuracy from ~70% to ~98%.
- Architected and developed WEnergyDocsEL, a high-performance async Python ETL pipeline that extracts, downloads, and loads financial documents from the WEnergy portal to Azure Blob Storage and Snowflake — achieving 68x faster processing than the previous sequential version.
- Implemented security-first design with Azure Key Vault integration, custom exception hierarchy (15+ types), and retry logic with exponential backoff for production-grade reliability.
- Streamlined and automated data collection for budgets, revenue forecasts, and customer targets using Python and PostgreSQL, enabling detailed performance tracking by sales office, route, SKU, and daily metrics.
- Developed and maintained Power BI dashboards for Sales, Finance, and HR functions, improving data accuracy and accessibility for critical business operations.
- Delivered training sessions for Raisa Energy engineers on AI coding assistants and modern development workflows, fostering a culture of AI-augmented productivity.

Tech: Snowflake, DBT, Python (async/await), Azure Blob Storage, Azure Key Vault, PostgreSQL, Power BI, SQL Server

Business Insights Senior Lead

Jan 2019 – Dec 2021

_VOIS (Vodafone)

Smart Village, Giza

- Engineered a Python desktop application (PyQT6 + Selenium + Win32com API) to automate recurring operational processes, reducing manual intervention by 25% and aligning with stakeholder requirements for intuitive UX.
- Owned and maintained multiple Power BI dashboards for Pre-sales, Demand & Utilization, and Work in Progress analysis, establishing data governance frameworks that ensured accuracy, integrity, and security.
- Collaborated with cross-departmental teams to embed analytics-driven strategies within organizational workflows, directly improving decision-making efficiency.
- Supported design and maintenance of a scalable Oracle SQL Data Warehouse, ensuring reliable and efficient data management for enterprise reporting.
- Executed data migration from on-premises systems to AWS Redshift and Google BigQuery, enhancing scalability and query performance for growing analytical workloads.

Tech: Python, PyQt6, Selenium, Power BI, Oracle SQL, AWS Redshift, Google BigQuery, Win32com API

Senior Business Intelligence Developer

Dec 2017 – Jan 2019

Atos

Cairo, Egypt (Hybrid)

- Served as technical lead for pre-sales activities on Enterprise BI and data platform projects, contributing to new client acquisition and relationship expansion through demonstrated technical expertise.
- Implemented MongoDB as Infrastructure-as-a-Service (IaaS) on Azure, deploying scalable and resilient databases for multiple environments and improving data storage reliability.
- Built an Elastic Cloud on Kubernetes (ECK) cluster on AKS for application resource monitoring, ensuring optimal resource utilization across Azure services.
- Enhanced security by integrating Azure authentication mechanisms into custom Python packages, enabling secure access to hybrid on-premises and cloud resources.
- Acted as primary analyst for the Microsoft Practice Team, delivering Power BI, Excel, and SQL Server reporting solutions that drove measurable business results.

Tech: Azure, MongoDB, Kubernetes (AKS/ECK), Power BI, Python, SQL Server, Excel

Data Analyst

May 2015 – Dec 2017

AI Safy Group, New Cairo

KEY PROJECTS

Competitor AI Insight – Agentic AI Pipeline (ITWorx)

Jan 2025 – Present

- Designed and built a POC Agentic AI system that automatically discovers, scrapes, and analyzes competitor intelligence across news, press releases, job postings, and RSS feeds from 28 sources — replacing manual monitoring with an AI-powered pipeline.

- Orchestrated 4 N8N workflows with Azure OpenAI GPT for structured extraction (13 fields per article), 6-topic classification with confidence scoring, sentiment analysis, and multi-engine scraping (HTTP, FlareSolvrr, Playwright, newspaper4k).
 - Engineered a star-schema Azure SQL data warehouse and Docker Compose infrastructure (N8N, PostgreSQL, Playwright scraper, FlareSolvrr, Caddy) — currently tracking 10 competitors with zero-code extensibility.
- Tech: N8N, Azure OpenAI GPT, Python, FastAPI, Playwright, Docker Compose, Azure SQL, PostgreSQL, LangChain*

Core42 – ADX Digital Transformation (ITWorx)

Jan 2024 – Dec 2024

- Led data engineering workstream for the Abu Dhabi Securities Exchange digital modernization, upgrading trading platforms and enhancing market accessibility for diverse stakeholders including investors, regulators, and clearing firms.
- Migrated and standardized 190+ Power BI Paginated Reports across 4 environments; built Python CLI to automate report styling (4hrs → 15min per batch).

Tech: Azure Synapse Analytics, Power BI Paginated Reports, Azure SQL Database, PostgreSQL, Python, PowerShell

Abraxes + MED@P Workflow Automation (ITWorx)

Oct 2023 – Jan 2024

- Developed a fully automated multilingual (German, French, Italian) workflow system for patient request handling across clinics, with data anonymization, automated clinic selection, and SLA-based penalty tracking.
- Built SSIS data pipelines and SSRS reports to streamline operations, reducing manual effort in public list report generation.

Tech: SSIS, SSRS, SQL Server, Visual Studio

Entity Correction Model (Raisa Energy)

Feb 2023 – Sep 2023

- Designed a multi-layer DBT model in Snowflake that automatically corrects revenue/expense entity allocations using exact and fuzzy matching against Jarvis ownership data, improving accuracy from ~70% to ~98%.
- Implemented a 4-status matching flow (Matched/Fixed/Multiple/No Match) that handles ~98.5% of records automatically, with only ~1.5% requiring manual review.

Tech: Snowflake, DBT, SQL

WEnergyDocsEL – Document ETL Pipeline (Raisa Energy)

Mar 2022 – Nov 2022

- Built an end-to-end async Python ETL pipeline that scrapes, downloads (64 concurrent), uploads (12 concurrent), and loads financial documents into Azure Blob Storage and Snowflake, achieving a 68x speedup over the previous sequential approach.
- Engineered production-grade reliability with Azure Key Vault credential management, 15+ custom exception types, and configurable retry logic with exponential backoff.

Tech: Python (asyncio), Azure Blob Storage, Azure Key Vault, Snowflake

CERTIFICATIONS

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- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
 - Microsoft Certified: Power BI Data Analyst Associate (PL-300)
 - Microsoft Azure Fundamentals (AZ-900)
 - Azure Data Engineering Associate (DP-203) – Coursera
 - Databricks Academy: SQL Analyst Associate
 - Google Cloud Data Engineer Certificate – Coursera
 - Google IT Automation with Python – Coursera
 - Data Engineering Nanodegree – Udacity
 - Data Streaming Nanodegree – Udacity
 - Cloud DevOps Engineer Nanodegree – Udacity
 - Machine Learning Engineer Nanodegree – Udacity
 - AI Programming with Python Nanodegree – Udacity
 - Advanced Data Analysis Nanodegree – Udacity
 - Deep Learning Specialization – Coursera (DeepLearning.AI)
 - Mathematics for Machine Learning Specialization – Coursera

EDUCATION

Bachelor of Commerce
Ain Shams University, Cairo

2008 – 2011

LANGUAGES

Arabic (Native) | **English** (Excellent)